



WHAT'S COOKING? • WOMEN IN DATA - DATATHON 2026

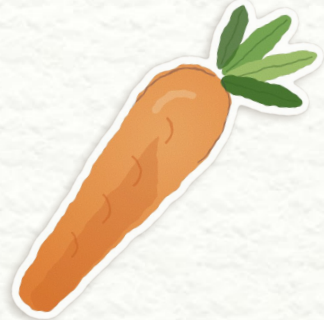
FROM FARM TO 400g

Can our food systems supply enough
fruits and vegetables for a healthy plate?

EAT • GROW • FAO • WHO

400 g / person / day • healthy-diet reference

Presented by the Byte Crew





TEAM



Meet the Byte Crew

TRACK: EAT x GROW



Linda Eva

Applied nutrition researcher turned business analytics specialist.



Dawn

Computer Science student and aspiring Data Engineer, transforming raw data into analysis-ready datasets.



Imane

Data Analyst



Laxmi

Business/Data Analyst



Monica

Operations and Insights Analyst for a nutrition company, specializing in voice of consumer analysis.





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PROBLEM STATEMENT & OBJECTIVES

Setting the table: our question, our objective, our stakeholders



Revealing global fruits and vegetables availability gaps, patterns, and opportunities for action.

400

g / person / day

WHO healthy-diet reference for fruits and vegetables.
Adequate intake is associated with **lower risk of cardiovascular disease and some cancers.**

USED AS A SCREENING BENCHMARK, NOT A VERDICT

Availability ≠ consumption · Correlation ≠ causation



OUR QUESTION

Can global supply reach the healthy plate?

Objective: measure the gap between national fruit-and-vegetable availability and 400 g/person/day, then investigate the food-system patterns behind the countries that fall short.



THREE RESEARCH QUESTIONS

Where? · Progress? · Patterns?

RQ1: where availability fell below 400.
RQ2: how countries moved from their first to their last observation.
RQ3: what distinguishes the countries still below the line.



WHO IT IS FOR

Public-health planners and policymakers

Nutrition and public-health bodies, agriculture and trade policymakers, FAO and WHO analysts: a screening signal that says where to look first.





What did we measure, and how?

A decade of FAOSTAT Food Balances, translated into grams on the plate.

THE DATA | FAOSTAT FOOD BALANCES

Source	FAOSTAT Food Balances (FAO); annual data, 2014–2023.
Coverage	178 countries/areas, 1,731 country-years; 168 complete, 10 partial.
Food groups	Fruits (excluding wine) + Vegetables.
Measure	Food supply quantity, kg/person/year, converted to g/person/day.
Calculation	$F\&V = \text{kg/person/year} \times 1,000 \div 365$; gap = availability – 400; adequacy % = availability $\div$ 400 $\times$ 100.
Supporting data	Production, imports and recorded loss share (RQ3).
Guardrails	National availability, not consumption; World and regions kept apart.

THE PIPELINE | BRONZE → SILVER → GOLD

1

BRONZE

Raw FAOSTAT extracts as downloaded: the two food groups and every supply element.

2

SILVER

Curated 2014–2023 records; source flags kept (E = estimated); aggregates and the aggregate China record kept out.

3

GOLD

One country-year table: g/person/day, gap to 400, benchmark and progress fields, per-person production, imports and loss share.

TOOLS

Data accessed from FAOSTAT; cleaned and analysed in Python (PySpark, pandas, NumPy) on Databricks; visualised with Matplotlib and Plotly.

OBSTACLES AND HOW WE ADDRESSED THEM

China aggregate overlap

The aggregate record was excluded; 178 countries/areas analysed instead of 179.

Uneven year coverage

10 of 178 countries have partial years: no imputation, available years used, headline re-checked on the 168 complete.

Structural missingness

Absent records were never set to zero; combined F&V only when both components exist; each indicator reports its own n.





Which parts of the world fall below the 400g benchmark?

Supply gaps cluster in parts of Africa, South and South-East Asia, and the Pacific, helping identify where food-system support may be most needed.

Period-Average F&V Availability by Country, 2014–2023



BELOW 400

57 of 178

countries (32.0%) were below 400 on period average.

WHAT THE MAP SHOWS

A substantial but uneven gap, concentrated in parts of Africa, South and South-East Asia and the Pacific.

GUARD RAIL

Country-level

Period-average classification; not population-weighted and not individual intake.



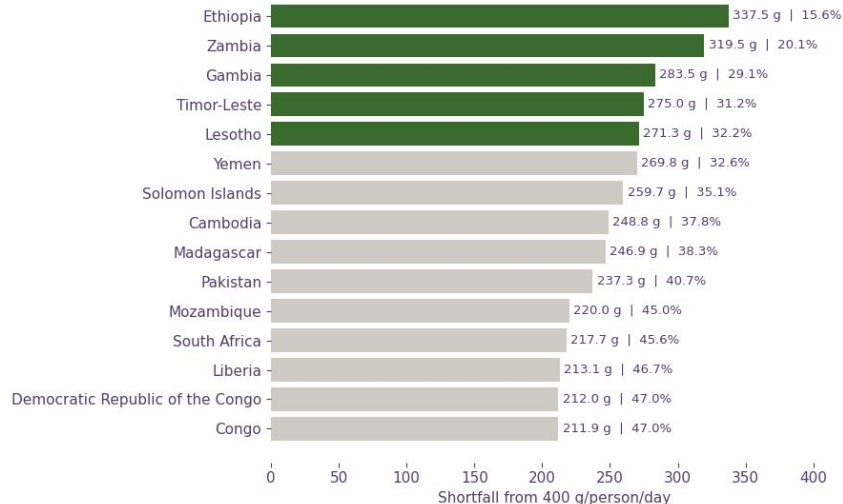


Which countries face the largest fruit-and-vegetable availability gap?

Ethiopia leads the deepest country-related shortfalls revealing where strengthening fruit-and-vegetable supply could have the greatest nutrition and food-security relevance.

Countries with the Largest Period-Average F&V Shortfalls, 2014–2023

Labels show shortfall (g/person/day) and Supply Adequacy %; top five highlighted



LARGEST GAP

Ethiopia

337.5 g/day below the reference; 15.6% supply adequacy.

NEXT LARGEST

Zambia • Gambia

Timor-Leste and Lesotho complete the five deepest shortfalls.

WHAT THE RANKING SHOWS

Gap size only

This is a size-of-gap list, not a to-do list. A country lower down may be improving; one higher up may already be closing its gap.

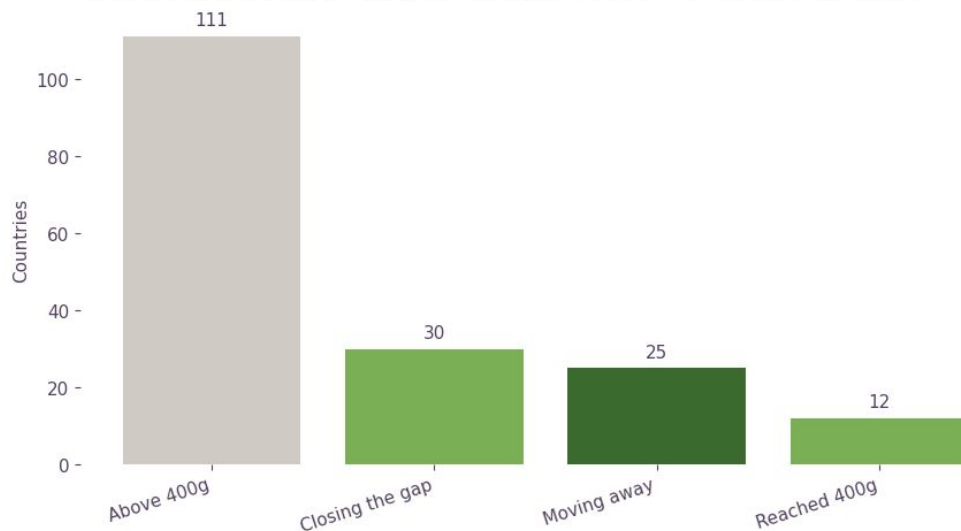




Are countries moving toward or away from the 400 g/day reference overtime?

Most countries remain above the benchmark, but diverging trajectories show why progress and emerging supply risks need to be tracked separately.

Benchmark-Transition Categories Relative to the 400 g/day Reference



BENCHMARK VIEW

111 • 12 • 30 • 25

Above 400 • Reached 400 • Closing the gap • Moving away

WHAT THE CHART SHOWS

Of 178 countries, 111 stayed above 400 across the period, 12 crossed the line to reach it, 30 are closing the gap, and 25 are moving away. No country ended exactly where it started.

WHAT IT MEANS

Benchmark shows each country's position relative to 400. It captures where they stand, not how fast they are moving within their category.



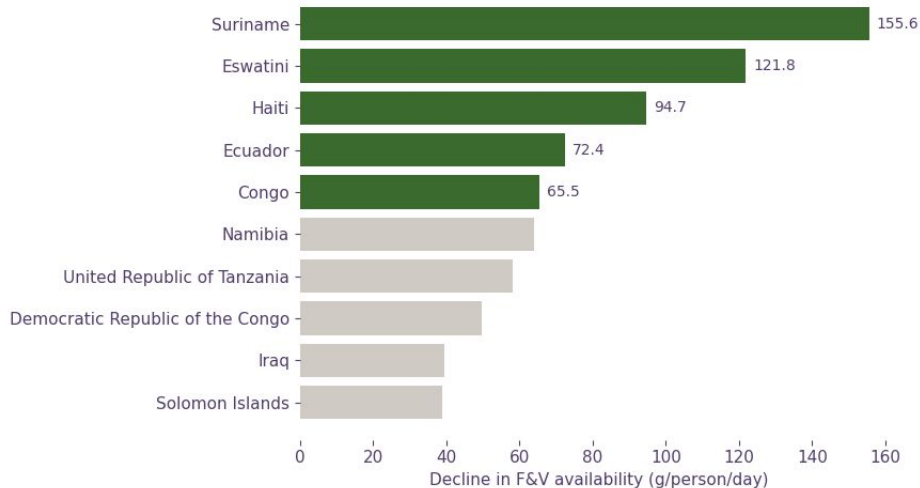


Which countries are moving away from the 400 g/day reference overtime?

Declining availability can act as an early warning for weakening food-system resilience before shortages become more severe.

Largest Declines Within the Hierarchical Moving Away Category

Top 10 within this category; the five largest are highlighted



TOP DECLINE

Suriname

155.6 g/person/day first-to-last decline

NEXT LARGEST

Eswatini • Haiti • Ecuador

Congo completes the five highlighted declines.

CRITICAL CAVEAT

Moving away ≠ all declines

55 countries declined while remaining Above 400 at both endpoints.





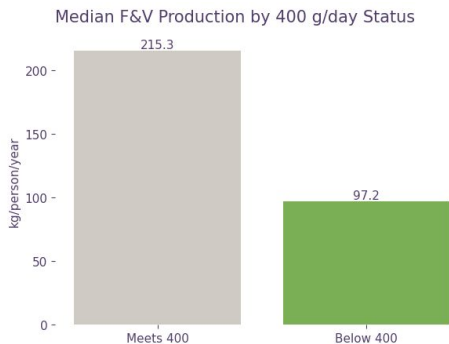
What food-system patterns distinguish countries below 400 g/day?

Lower production and import levels point to both domestic supply and trade resilience as important areas to further investigate.

PRODUCTION

97.2 vs 215.3

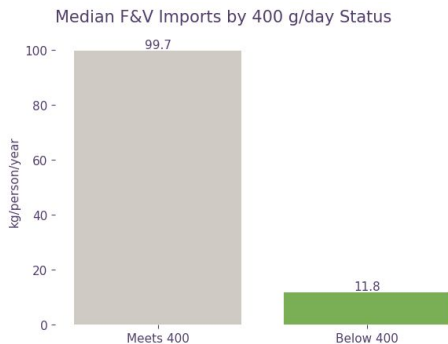
Below-400 countries showed markedly lower production capacity



IMPORTS

11.8 vs 99.7

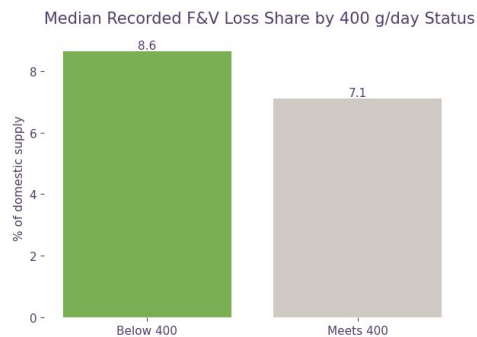
Below-400 countries also showed much lower importation levels



RECORDED FOOD LOSSES

8.6% vs 7.1%

Recorded food loss levels looked fairly similar in both groups



Available-case medians. n varies by indicator: production 56/120; imports 57/121; loss share 57/120.





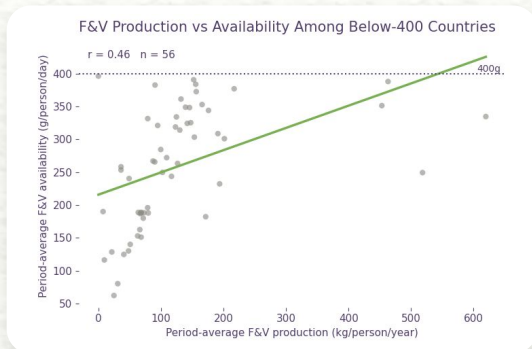
Which food-system indicators are most closely associated with availability?

Production shows the strongest signal, suggesting that sustainable production may be an important lever, while trade could still help complement local supply.

PRODUCTION

$$r = 0.46$$

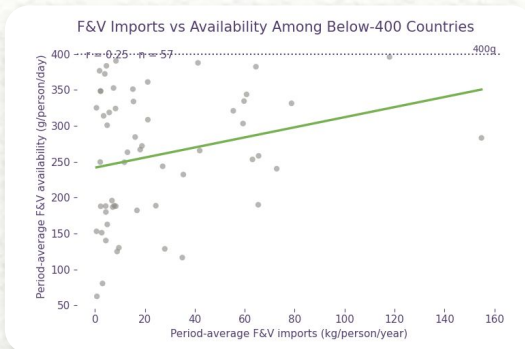
Moderate positive association • n = 56



IMPORTS

$$r = 0.25$$

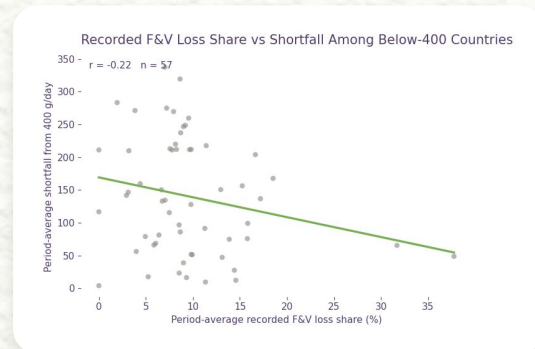
Weak positive association • n = 57



RECORDED FOOD LOSS

$$r = -0.22$$

Weak inverse link with shortfall • n = 57



ROBUSTNESS Spearman $\rho = 0.65 / 0.18 / -0.19$ on the same samples; the 168 complete-coverage countries give 53 below 400 (31.5%) and the same 25 moving away.

CORRELATION $\neq$ CAUSATION Pearson r among below-400 countries. Production and imports are components of the Food Balance accounting structure, so no causal reading is made.





SOLUTION HIGHLIGHTS



A Scalable Approach to Identifying and Closing Fruit-and-Vegetable Supply Gaps



SCREEN

Compare every country's fruit and vegetable supply with 400g.



TRACK

Follow each country's direction year after year, not just its position.



DIAGNOSE

Evaluate food-system patterns like production or imports side by side.



ACT

Match the pattern to a possible response.

WHAT WE BUILT

A Python-based screening signal analysis that could evolve into a solution framework for identifying fruit-and-vegetable supply gaps, tracking progress over time, exploring food-system patterns, and linking findings to potential responses.

WHY IT MATTERS

It turns complex food-system data into a simple signal that can help countries identify where nutrition-related supply risks may be emerging, and where production, trade or post-harvest systems may need closer attention.

POTENTIAL TO SCALE

The screen could be refreshed annually as new data become available, and expanded with prices, access, affordability and household dietary surveys to connect national availability more closely with what people can actually obtain and eat.

ETHICAL CONSIDERATION

400 g is a screening signal, not a verdict. Availability does not equal consumption, and correlations do not establish causation; local context should guide any policy response.

Find the gap. Follow its direction. Investigate what may be driving it. Target resources wisely.





Which response fits the gap constraint a country might have?

Diagnose the main problem first, then choose the response that fits the country.



IF PRODUCTION IS LOW

1 Grow more fruit & vegetables

Support climate-resilient farming, small producers, practical farm advice, research and better links to markets. [1][2]

Expected effect: a larger, more reliable local supply.



IF LOCAL SUPPLY IS NOT ENOUGH

2 Use trade to help fill gaps

Combine local production with a wider range of import sources so supply stays reliable through the year. [2][3]

Expected effect: a steadier supply from more than one source.



IF FOOD IS LOST AFTER HARVEST

3 Improve storage & transport

Where losses are high, improve handling, storage, cold rooms, transport and market systems. [4][5]

Expected effect: more edible produce reaches markets.



EVEN ABOVE 400

4 Check price, access & diet

Track food prices, healthy-diet costs, access and household diets. [1][5]

Expected effect: food that people can actually obtain.

[1] WHO Healthy diet (2026) • [2] FAO Fruits & Vegetables: Take Action • [3] SOCO 2024 • [4] FAO Food Loss data (2025) • [5] SOFI 2026

Next research step: pair the 400 g availability screening signal study with household dietary surveys to test how closely national supply reflects what people actually eat or buy.





So, can our food systems supply enough fruits and vegetables for a healthy plate?

For one in three countries, a healthy plate made up of fruits and vegetables may depend on agriculture and trade before diet, while many others could face the challenge of preserving the food availability they already have.

SHORT OF 400 g ON AVERAGE

57 countries

The healthy plate could be an agriculture and trade question before it is a diet question.

ABOVE THE LINE, AND SLIPPING

55 countries

The question could be holding on to what they already have, before the line is crossed.

ABOVE THE LINE, AND STEADY

66 countries

Keep checking every year: where a country stands today is not where it could be heading tomorrow.

178 countries, 2014–2023, on one scale

IMPACT ON FOOD SYSTEMS AND OUR PLANET

Our work could evolve into an annual 400g screening solution framework that helps countries identify and close fruit-and-vegetable supply gaps while directing land, water, and funding resources more wisely to support healthier diets.



Thanks For Your Kind Attention!

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